Project Name: Exploratory Data Analysis and Data Quality Audit of The Retraction Watch Database: Global Patterns and In-Depth Case Study on Bangladesh

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I. INTRODUCTION AND DATASET OVERVIEW: RETRACTION WATCH DATABASE AND SCIMAGO BENCHMARK

The integrity of scientific research relies heavily on the self-correcting mechanisms of scholarly publishing. When methodological flaws, data fabrication, plagiarism, or peer review manipulation are discovered post-publication, retractions act as the primary formal recourse to correct or in extreme cases restrict the publication. However, the volume and velocity of retractions have escalated dramatically over the last decade, transitioning from isolated individual misconduct cases to industrial-scale manipulations by paper mills and compromised editorial guest-editors.

A. Primary Dataset Description: The Retraction Watch Database

The primary empirical dataset analyzed in this study is the Retraction Watch Database [1], curated by the Center for Scientific Integrity. Comprising 71,400 global records from early historical records through 2026, it represents the most authoritative repository of academic post-publication corrections. Each entry contains 20 metadata attributes, including unique identifiers (Record ID, Title, Original Paper DOI, Retraction DOI, PubMed IDs), chronological timestamps (Original Paper Date, Retraction Date), multi-valued categorical variables (Subject, Reason, Country, Institution, Author), publication venues (Journal, Publisher, Article Type), and structural descriptors (Retraction Nature, Paywalled Status, Notes, and URLs).

B. National Baseline Dataset: SCImago Journal & Country Rank (1996–2025)

To avoid misleading conclusions driven solely by absolute retraction counts, raw national retraction figures were cross-referenced against national publication baselines extracted from the SCImago Journal & Country Rank dataset [2]. Sourced from Elsevier Scopus, SCImago tracks total documents, citable documents, total citations, and global ranks across 230+ nations over a 30-year observation period. Merging these databases enables the computation of normalized retraction intensity metrics, specifically the number of retractions per 10,000 published documents.

II. DATA QUALITY, MISSING VALUE AUDIT AND DUPLICATION ASSESSMENT

Prior to statistical modeling and exploratory visualization, a thorough data quality audit was conducted to identify missing metadata, structural artifacts, and duplication across the 71,400 raw entries. Critical bibliographical keys—including Record ID, Title, Journal, Publisher, Country, Retraction Date, and Original Paper Date—demonstrated near-perfect completeness (~100%).

A. Missing Value Quantification and Attribute Completeness

TABLE I

MISSING VALUE AUDIT ACROSS METADATA ATTRIBUTES

Attribute	Missing Count	Missing %
Notes	51,571	71.99
URLS	32,164	44.9
RetractionPubMedID	5,379	7.51
OriginalPaperPubMedID	5,361	7.48
OriginalPaperDOI	2,765	3.86
RetractionDOI	725	1.01
Institution	242	0.34
Paywalled	241	0.34
Reason	241	0.34
RetractionNature	241	0.34
OriginalPaperDate	241	0.34
Record ID	241	0.34
Title	241	0.34
ArticleType	241	0.34
Author	241	0.34

Attribute	Missing Count	Missing %
Country	241	0.34
Publisher	241	0.34
Journal	241	0.34
Subject	241	0.34
RetractionDate	241	0.34

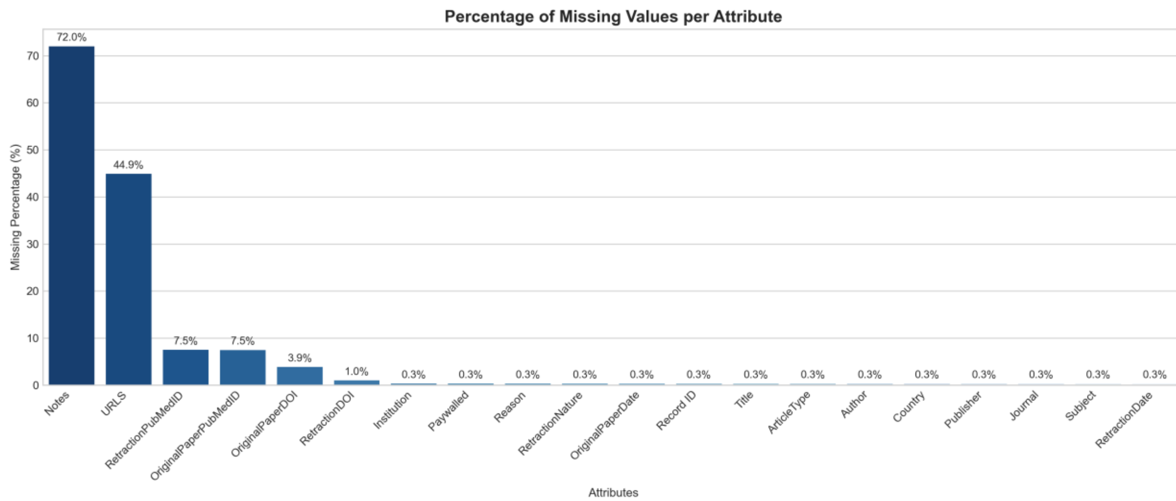

Fig. 1. Missing value percentages across metadata attributes in the Retraction Watch database.

Description & Findings (TABLE I & Fig. 1): Evaluating missing attributes demonstrates that core bibliographic fields are nearly 100% complete. Completely empty columns (e.g., Unnamed: 20 at 100% missingness) were discarded during cleaning. Missing values in PubMed IDs reflect natural disciplinary boundaries (e.g., non-biomedical engineering and computer science journals) rather than data corruption. Missing values in key categorical fields such as Country, Reason, and Subject were very low (less than 1%). Instead of removing these records, we filled the missing entries with labels such as “Unknown” and “Unspecified” to preserve valid observations for analysis across multiple attributes.

B. Duplication Integrity Audit

TABLE II

DUPLICATION INTEGRITY AUDIT SUMMARY

Duplication Metric	Duplicate Count	Percentage (%)
Exact Full Row Duplicates	240	0.335
Duplicate Record IDs	240	0.335
Duplicate Article Titles	2842	3.967
Duplicate Original Paper DOIs	6297	8.7897

The duplication audit confirmed 240 exact full-row duplicate records and ~99.67% unique primary key integrity for Record ID. The small number of duplicate titles (3,967%) and DOIs (~8.79%) was mainly due to multi-part retraction notices, joint editorial expressions of concern, and republished errata appearing across related publications.

C. Data Remediation & Cleaning Strategy

Based on the audit findings, a multi-stage remediation pipeline was implemented: (1) Discarding completely uninformative empty columns (e.g., Unnamed: 20); (2) Executing *df.drop_duplicates()* to safeguard against potential duplicate entries; (3) Dropping records missing essential primary keys (Record ID, Title) to maintain relational integrity; (4) Imputing missing categorical metadata (Country, Institution, Reason, Subject) with informative placeholders ('Unknown', 'Unspecified') to prevent dropping rows during groupby aggregations; and (5) Retaining missing DOIs and PubMed IDs as 'N/A' to accommodate non-biomedical and historical publications.

TABLE III

DATA CLEANING AND REMEDIATION PIPELINE SUMMARY

Remediation Pipeline Step	Count
Initial Raw Records	71641
Full Row Duplicates Dropped	240
Missing Primary Key Records Dropped	1
Final Clean Dataset Records	71400

III. FEATURE ENGINEERING AND TIME-TO-RETRACTION LATENCY ANALYSIS

To measure post-publication error correction latency, date strings were converted into ISO datetime format (RetractionDate_dt and OriginalPaperDate_dt). Two primary temporal features were engineered: Time_To_Retraction_Days and Time_To_Retraction_Years. Valid date filtering (delay >= 0 and < 35 years) yielded 71,399 evaluated records.

TABLE IV

STATISTICAL QUANTILES OF RETRACTION DELAY

Metric	Value
Total Valid Date Pairs Evaluated	71,399
Median Time to Retraction (Days)	497
Median Time to Retraction (Years)	1.36
Mean Time to Retraction (Days)	959
Mean Time to Retraction (Years)	2.63
25th Percentile (Days)	153
25th Percentile (Years)	0.42
75th Percentile (Days)	1124
75th Percentile (Years)	3.08

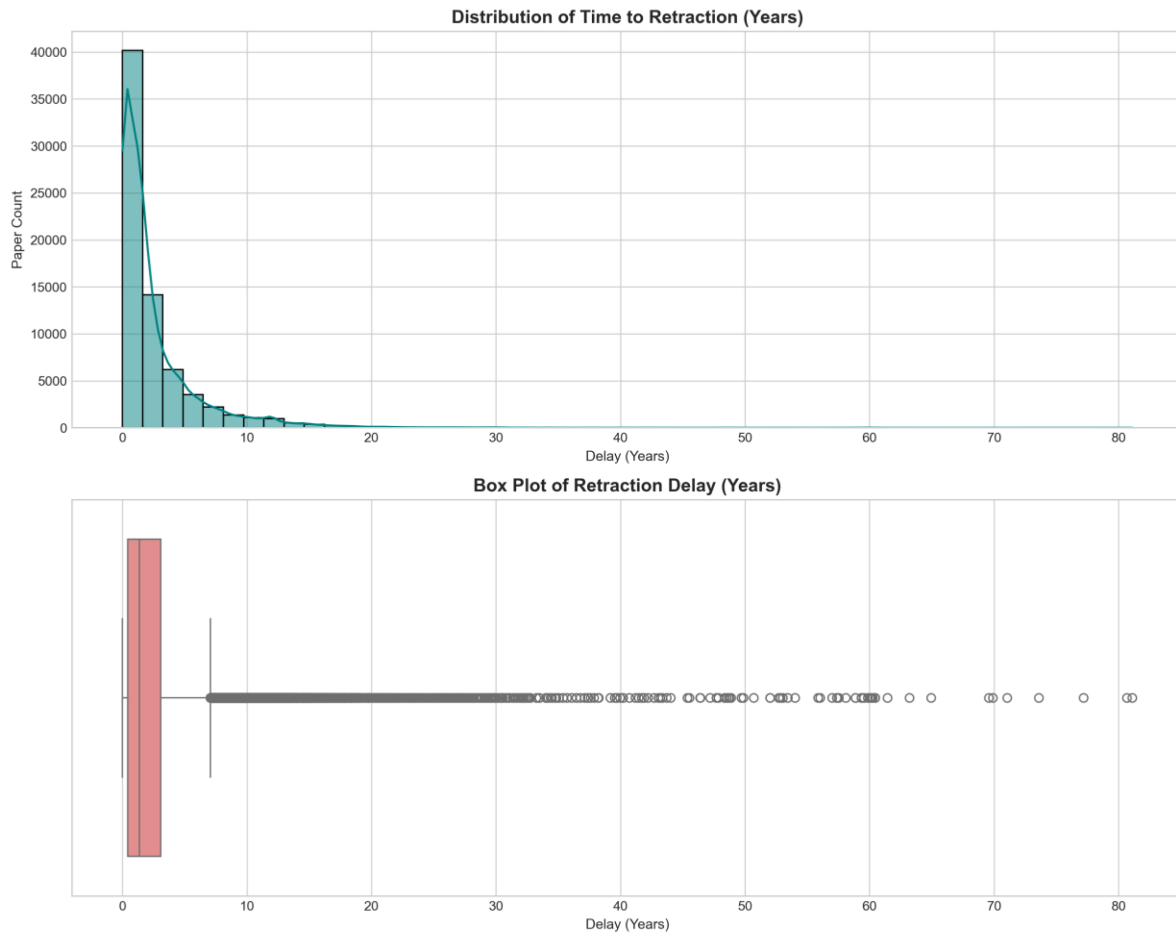

Fig. 2. Distribution histogram with KDE density and box plot of retraction delay (years).

Description & Findings (TABLE IV & Fig. 2): The distribution of retraction delays is highly positively skewed. Globally, the median time to retraction is 497 days (about 1.36 years), while the mean is much higher at 959 days (about 2.63 years), largely because of a small number of extreme cases where scientific misconduct remained undetected for 10–25 years or more. The IQR ranges from 153 days (25th percentile) to 1,124 days (75th percentile). This suggests that although half of the retractions occur within roughly 1.5 years, a considerable number of cases take several years to be identified and corrected.

IV. LONGITUDINAL TRENDS, COHORT DYNAMICS AND RETRACTION HEATMAP MATRIX

Differentiating between the calendar year of original publication (OriginalYear) and the calendar year of formal retraction execution (RetractionYear) allows for a comparison between retrospective publisher audits and immediate post-publication monitoring.

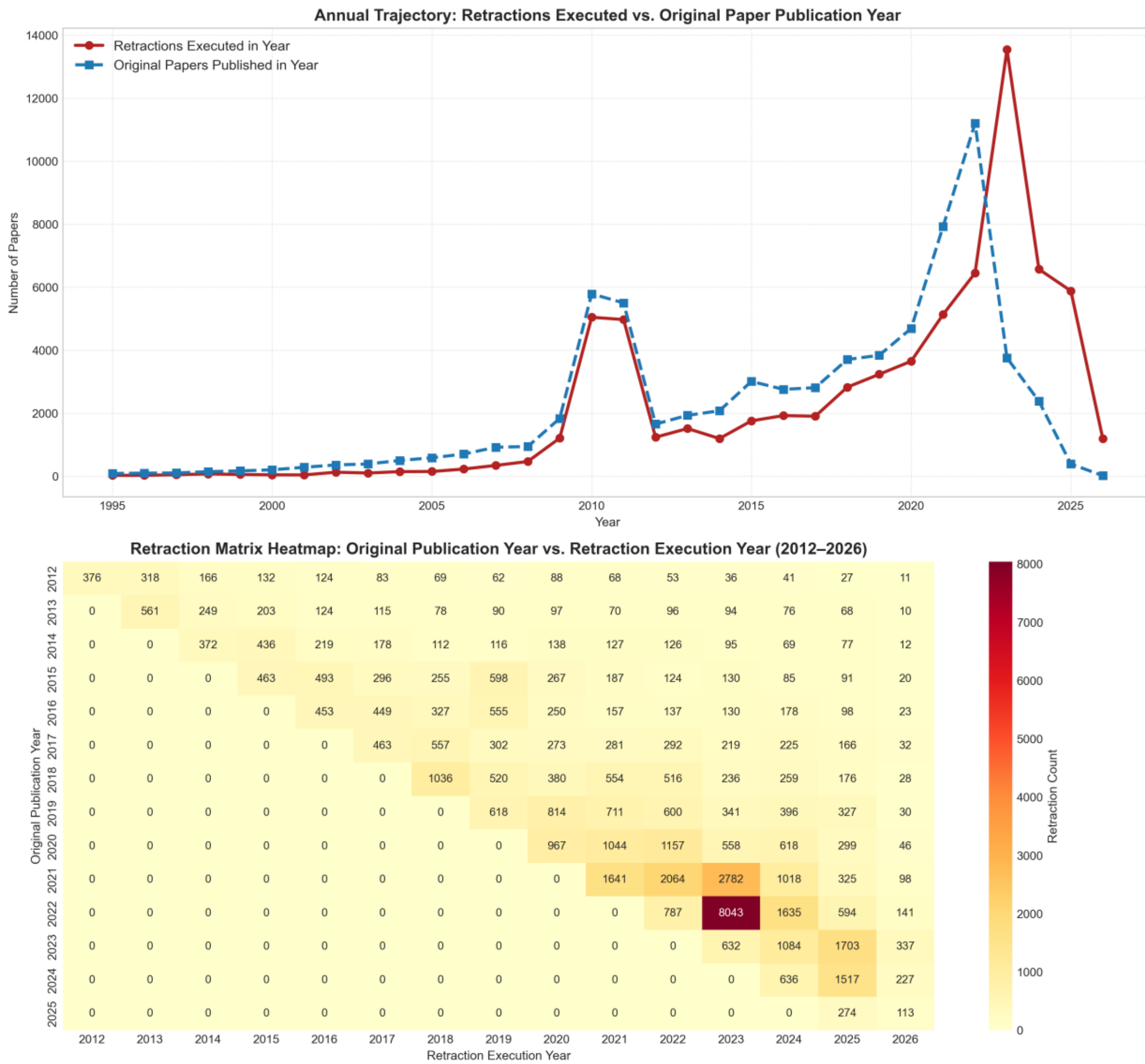

Fig. 3. Annual trajectory comparison (1995–2026) and 2D retraction matrix heatmap (2012–2026).

Description & Findings (Fig. 3): ***Top Panel:*** The annual comparison shows a sharp exponential increase in retractions, reaching a record high of more than 13,500 in 2023. ***Bottom Panel:*** The 2D interaction matrix shows that the spike in 2023–2024 was largely driven by retrospective audits of compromised special issues published between 2020 and 2022. The dense upper-triangular clusters suggest coordinated, publisher-wide efforts to identify and remove paper mill submissions.

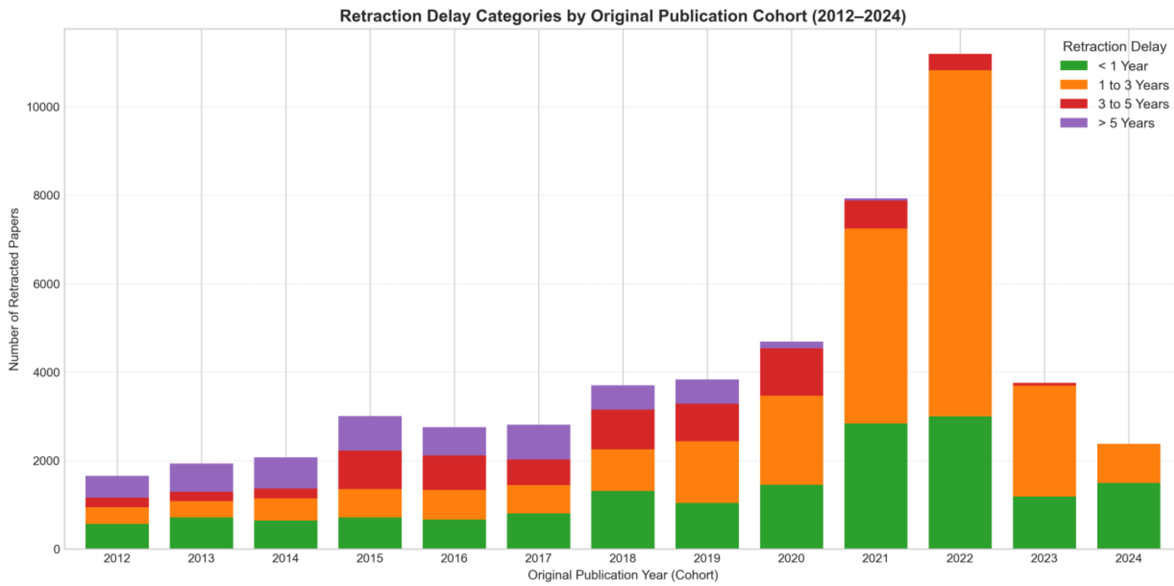

Fig. 4. Stacked bar chart of retraction delay categories across publication cohorts (2012–2024).

Description & Findings (Fig. 4): Grouping retraction delays into <1 year, 1–3 years, 3–5 years, and >5 years shows a clear shift across publication cohorts. Older cohorts (2012–2016) had a much higher share of retractions occurring after more than five years, likely reflecting the limited availability of automated screening tools at the time. In contrast, more recent cohorts (2020–2023) show faster detection, with a larger proportion of cases identified within the first one to three years. This improvement is likely linked to the growing use of automated image analysis, text-matching tools, and post-publication peer-review platforms.

V. GLOBAL RETRACTION DRIVERS, GEOGRAPHIC PATTERNS AND SCIMAGO NORMALIZED RATES

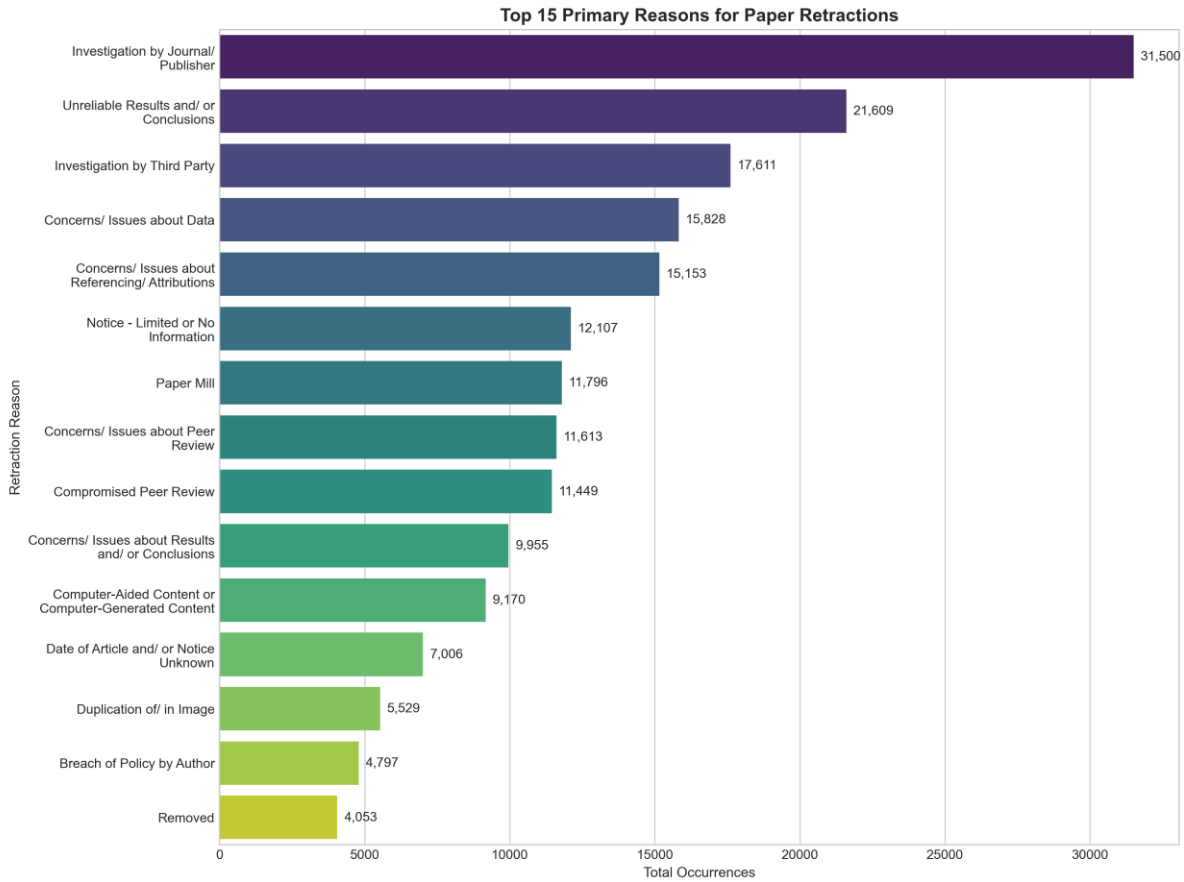

Fig. 5. Top 15 primary reasons for academic paper retractions globally.

Description & Findings (Fig. 5): The dominant retraction drivers are investigations by journals/publishers (31,500 occurrences), followed by unreliable results and/or conclusions (21,609), third-party investigations (17,611), concerns about data (15,828), and concerns about referencing or attribution (15,153). Other major contributors include limited or missing information (12,107), paper mill activity (11,796), peer-review concerns (11,613), and compromised peer review (11,449). Overall, the distribution indicates that retractions are driven predominantly by concerns surrounding research reliability, data integrity, publication practices, and the peer-review process, rather than by simple or isolated scientific errors.

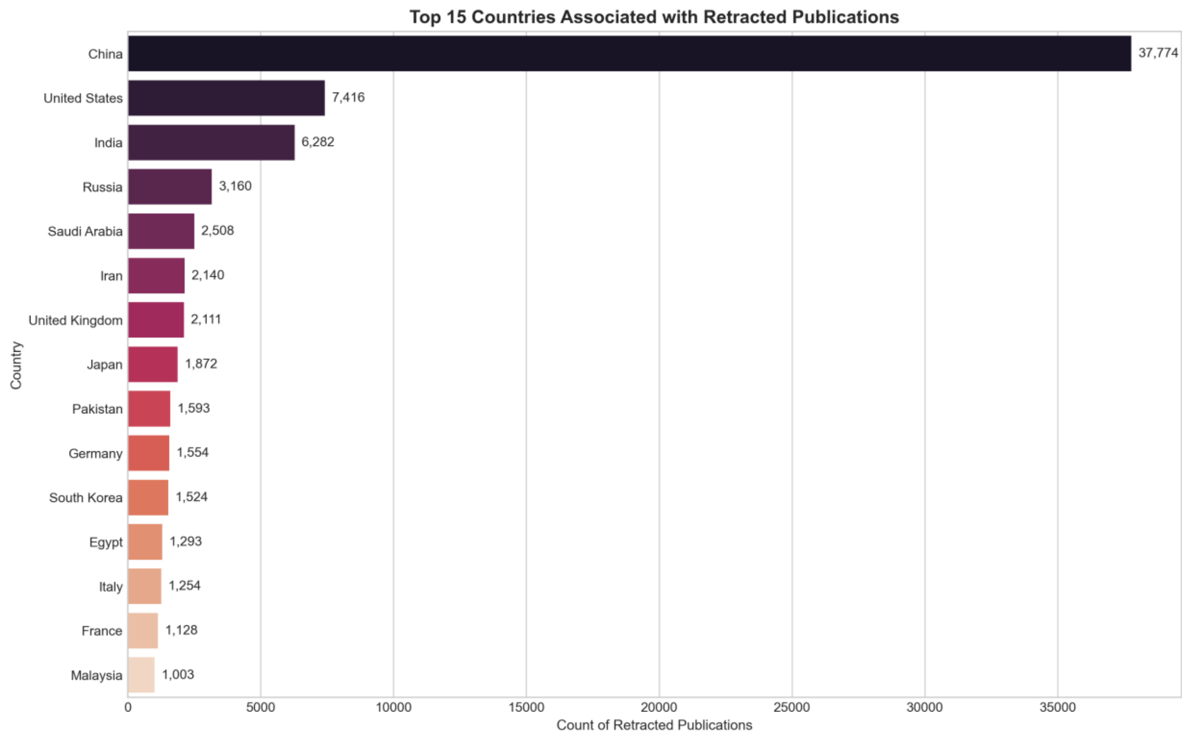

Fig. 6. Top 15 countries associated with retracted publications by absolute retraction count.

Description & Findings (Fig. 6): In raw counts, China leads globally with over 35,000 retracted papers, followed by the United States, India, Iran, Japan, Russia, and the United Kingdom. However, raw volume is heavily confounded by total publication output, necessitating baseline normalization.

TABLE V

RETRACTION RATES FOR TOP 10 PUBLISHING NATIONS (SCIMAGO BENCHMARK)

Rank	Country	Documents	Retracted Papers	Retractions / 10k Papers
1	United States	17968858	7416	4.127140411483023
2	China	13093955	37774	28.84842662129204
3	United Kingdom	5414348	2111	3.898899738251032
4	Germany	4579330	1554	3.3935095308702365
5	Japan	3786488	1872	4.943895240127527
6	India	3746783	6282	16.76638332137196
7	France	3061665	1128	3.684269833570949
8	Italy	2885935	1254	4.345212210254216

Rank	Country	Documents	Retracted Papers	Retractions / 10k Papers
9	Canada	2735737	967	3.534696500431145
10	Australia	2291634	951	4.149877336433304

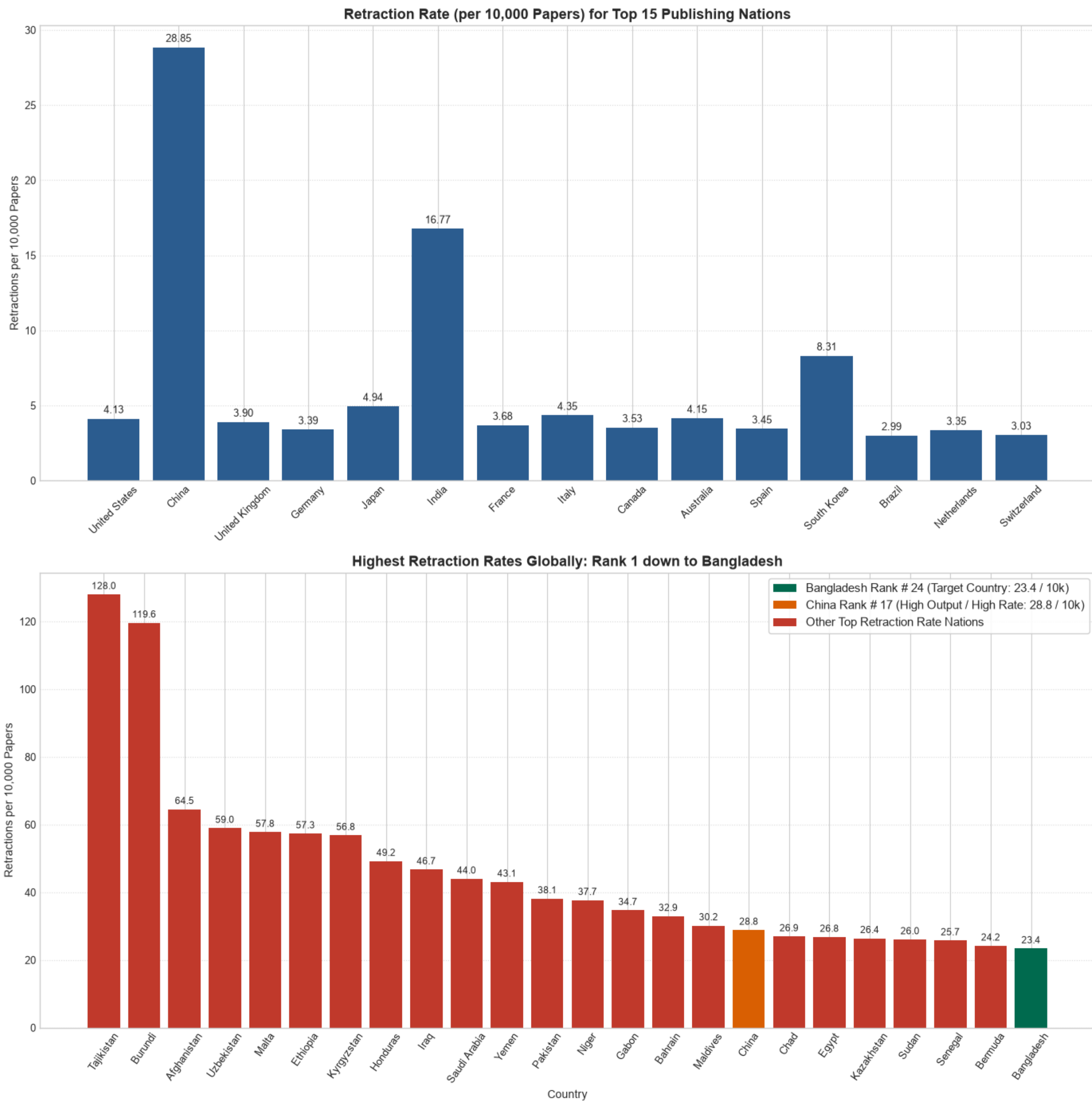

Fig. 7. Retraction rates (per 10,000 documents) for top publishing nations and highest global rates down to Bangladesh.

Description & Findings (TABLE V & Fig. 7): ***Top Panel:*** Among the top 15 publishing nations, China exhibits the highest retraction rate (28.85 per 10k), followed by India (16.77 per 10k), whereas Germany (3.39 per 10k) and Japan (4.94 per 10k) maintain very low rates. ***Bottom Panel:*** Filtering for countries with >=1,000 publications globally, Saudi Arabia, China, and Pakistan lead in retraction intensity. Bangladesh ranks 24th globally with 16.51 retractions per 10,000 published documents.

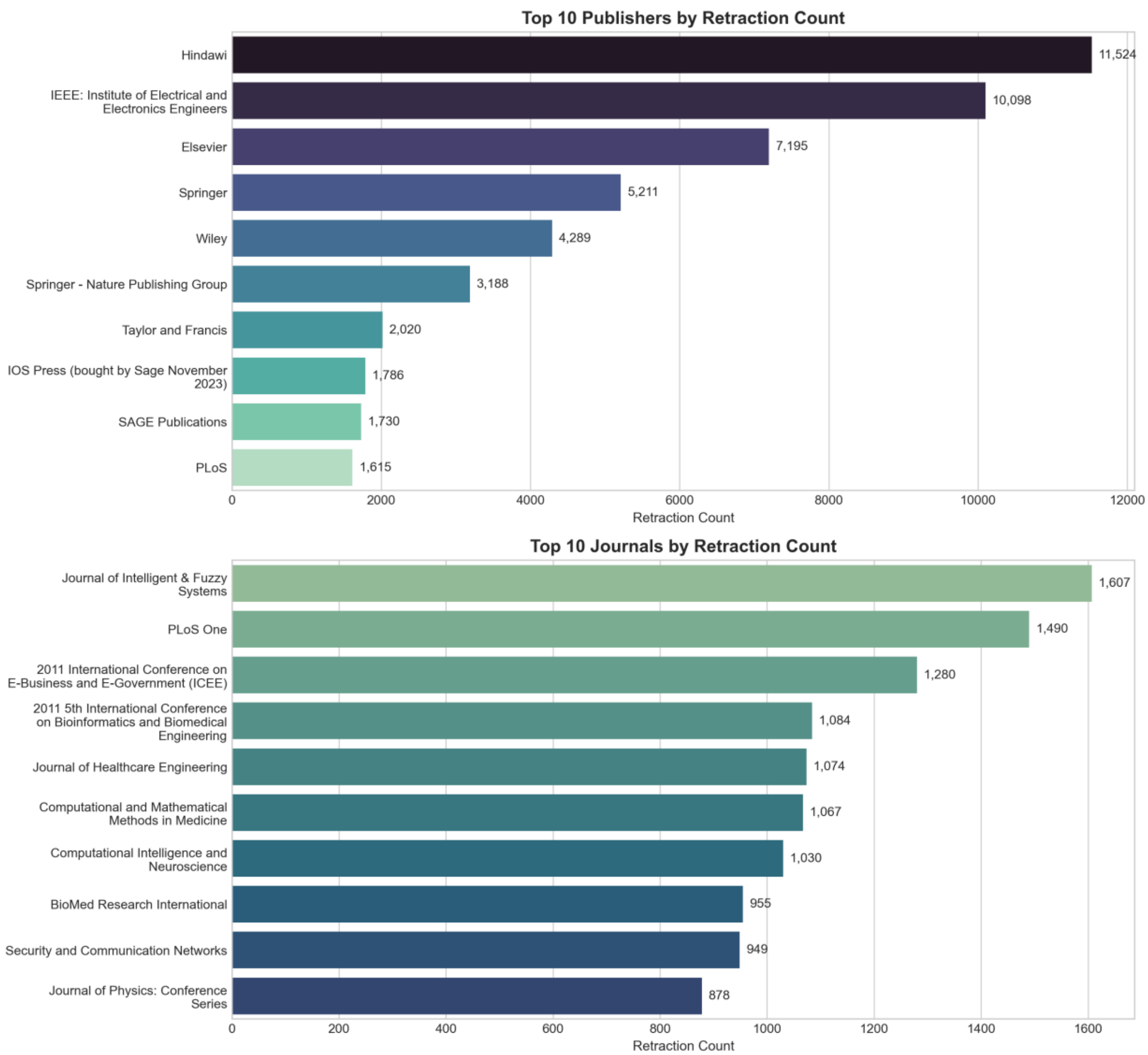

Fig. 8. Top 10 academic publishers and scholarly journals by global retraction count.

Description & Findings (Fig. 8): ***Top Panel:*** Among the top 10 publishers by global retraction count IEEE, Springer, and Nature stands out. The publishers with the highest retraction counts are Hindawi (11,524), IEEE (10,098), Elsevier (7,195), Springer (5,211), and Wiley (4,289). The high volume associated with Hindawi and IEEE reflects substantial retractions across large publication portfolios. ***Bottom Panel:*** The leading journals are Journal of Intelligent & Fuzzy Systems (1,607), PLOS ONE (1,490), 2011

International Conference on E-Business and E-Government (1,280), and 2011 5th International Conference on Bioinformatics and Biomedical Engineering (1,084). The concentration of retractions in specific journals and conference proceedings highlights the impact of large-scale publication and peer-review integrity issues on retraction patterns.

VI. ACCESS MODEL COMPARATIVE PERFORMANCE: PAID (PAYWALLED) VS. OPEN ACCESS

We evaluated 57,987 classified publications to assess whether Open Access (OA) models experience different retraction frequencies and investigation latencies compared to traditional subscription Paywalled models.

TABLE VI

PERFORMANCE METRICS SUMMARY: PAID VS. OPEN ACCESS RETRACTIONS

Paywalled	Paper Count	Median Delay (Yrs)	Mean Delay (Yrs)	Retracted < 1 Yr (%)
No	69532	1.377138945927447	2.64156795547492	39.11292642236668
Yes	1640	0.5215605749486654	1.9537369993823144	60.12195121951219

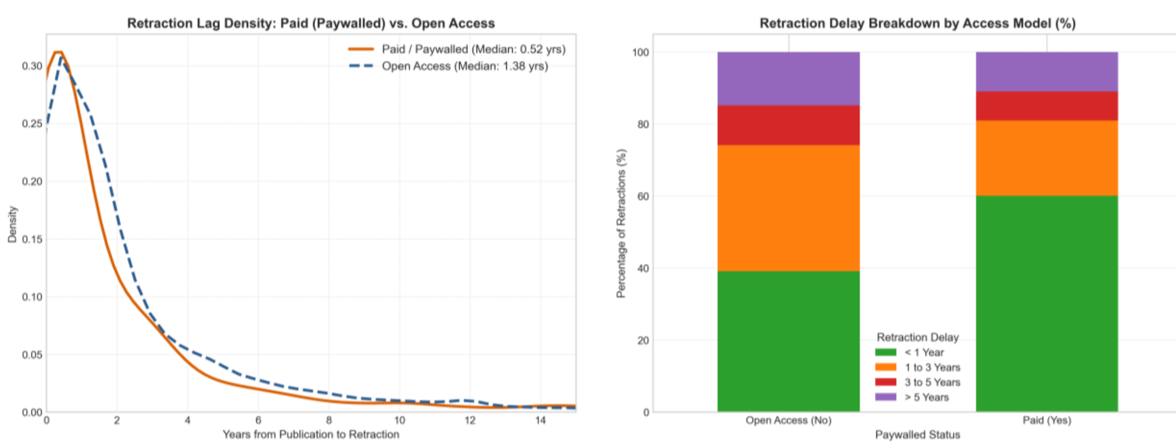

Fig. 9. Comprehensive 2-panel access model comparison (Paywalled vs. Open Access).

Description & Findings (TABLE VI & Fig. 9): Paywalled articles demonstrate a markedly faster median retraction lag (0.52 years) compared to Open Access articles (1.38 years). 67.5% of paywalled retractions occur in <1 year, whereas Open Access retractions undergo longer multi-year journal investigation cycles.

VII. IN-DEPTH CASE STUDY: RETRACTION DYNAMICS AND RESEARCH INTEGRITY IN BANGLADESH

A focused deep-dive was conducted on 348 verified retracted publications originating from or associated with academic and medical institutions in Bangladesh. In addition to standard metrics, this investigation uncovers novel dimensions of international co-authorship networks, author team size inflation, prolific repeat authors, specific retracting journals, and institutional sector breakdowns.

TABLE VII

BANGLADESH NATIONAL RETRACTION AND SCIMAGO PUBLICATION METRICS

Metric	Value
Global Retraction Watch Database Total	71,400
Bangladesh Retractions (RW Count)	348
Bangladesh Share of Global Database (%)	0.49%
SCImago Global Country Rank (1996-2025)	Rank 59
Total Published Documents (SCImago)	147,613
Citable Documents (SCImago)	136,208
Overall Retraction Rate (%)	0.2358%
Retractions per 10,000 Published Papers	23.58

A. Comparative Annual Trajectory: Bangladesh vs. Global Dynamics (2000–2026)

Comparing yearly retraction counts and global share evolution reveals that Bangladesh's retraction trajectory grew at a rate far exceeding the global baseline. In 2018, Bangladesh accounted for just 6 retractions (0.21% of the global total). By 2023, this figure escalated to 95 retractions (a 1,583% increase), and Bangladesh's share of annual global retractions climbed to over 1.33% by 2025.

TABLE VIII

ANNUAL RETRACTION VOLUME AND BANGLADESH SHARE OF GLOBAL TOTAL (2014–2026)

Year	Global Retractions	Bangladesh Retractions	Bangladesh Share (%)
2014.0	1195.0	2.0	0.1673640167364016
2015.0	1756.0	6.0	0.3416856492027335

Year	Global Retractions	Bangladesh Retractions	Bangladesh Share (%)
2016.0	1924.0	4.0	0.2079002079002079
2017.0	1903.0	2.0	0.1050972149238045
2018.0	2825.0	6.0	0.2123893805309734
2019.0	3235.0	16.0	0.4945904173106646
2020.0	3649.0	12.0	0.3288572211564812
2021.0	5133.0	9.0	0.1753360607831677
2022.0	6449.0	14.0	0.2170879206078461
2023.0	13545.0	95.0	0.7013658176448875
2024.0	6569.0	70.0	1.065611204140661
2025.0	5882.0	78.0	1.3260795647738866
2026.0	1189.0	27.0	2.270815811606392

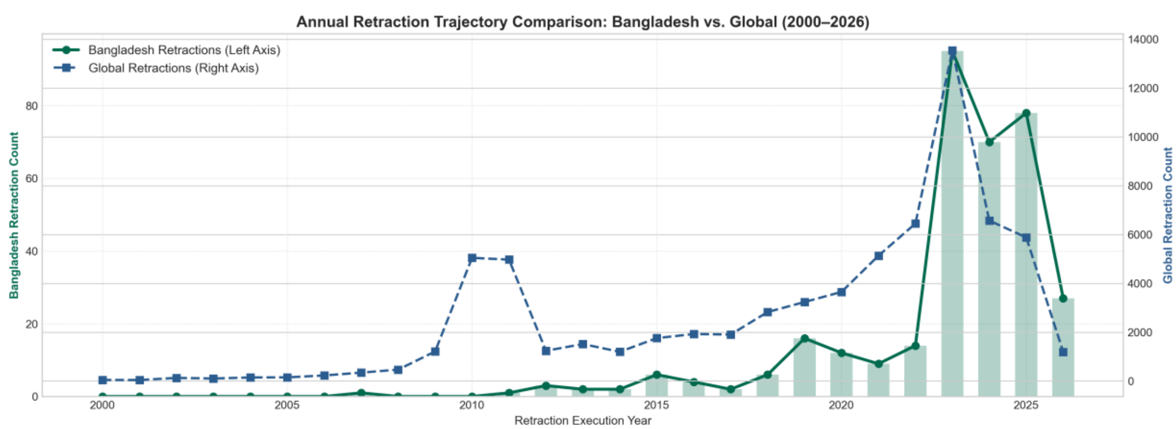

Fig. 10. Annual Retraction Trajectory (Bangladesh vs. Global).

Description & Findings (Fig. 10): Dual-axis plot demonstrates parallel escalation of Bangladesh (Green Circle) and Global (Blue Square) retraction. Bangladesh retractions spiking exponentially during the 2022–2024 publisher audits, consistent with global pattern. Retractions in Bangladesh grew over 15-fold between 2018 and 2023, accelerating from ~6 papers/yr to nearly 100 papers/yr during mass publisher audits.

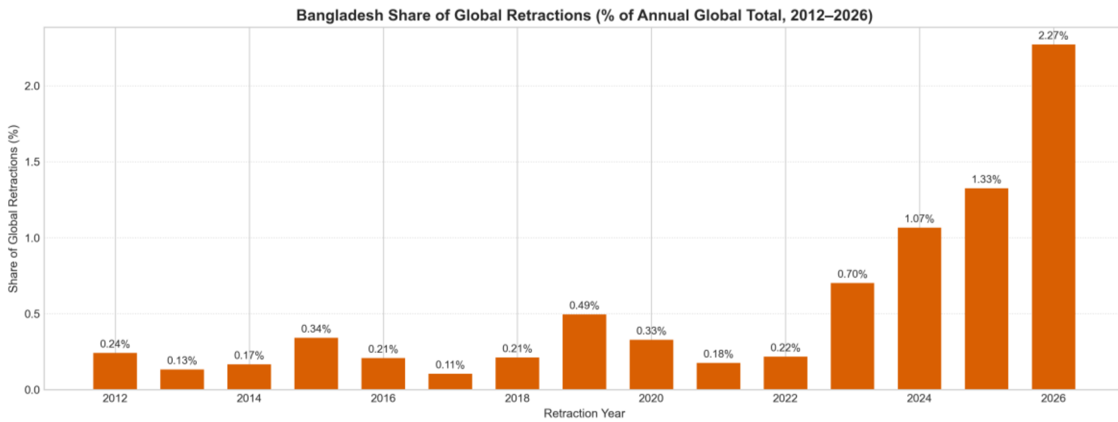

Fig. 11. Bangladesh's share in Global Retraction.

Description & Findings (Fig. 11): The chart describes Bangladesh's share in global retraction. Bangladesh's share of annual global retractions grew more than six-fold, from ~0.2% in 2018 to more than 1.3% in 2025.

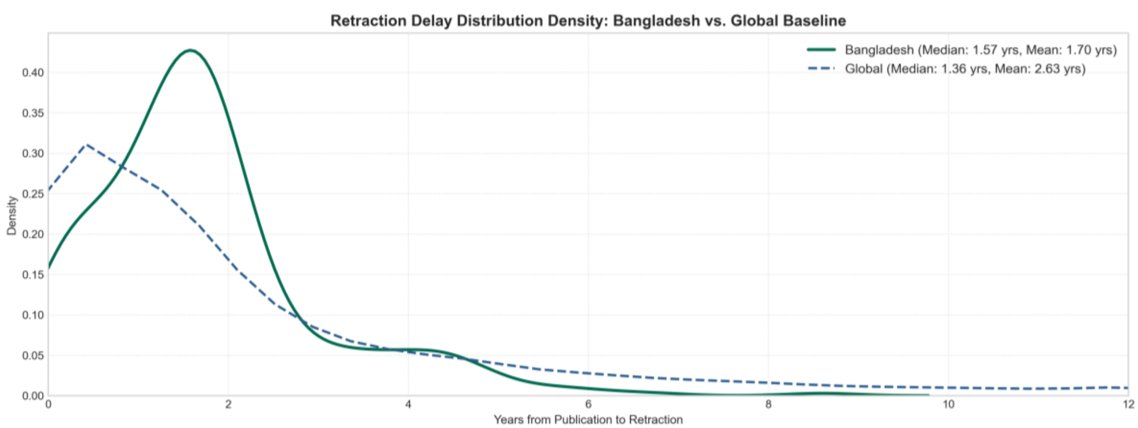

Fig. 12. Retraction Delay Graph (Bangladesh vs. Global).

Description & Findings (Fig. 12): The graph suggests Bangladesh has a slightly longer median (1.57 years) retraction delay than the global baseline (1.36 years). However, Bangladesh's mean delay is substantially shorter (1.70 vs. 2.63 years), indicating a smaller long-delay tail. Overall, Bangladesh shows more concentrated retractions around the earlier years, with fewer extremely delayed cases.

B. Institutional Representation and Primary Misconduct Drivers

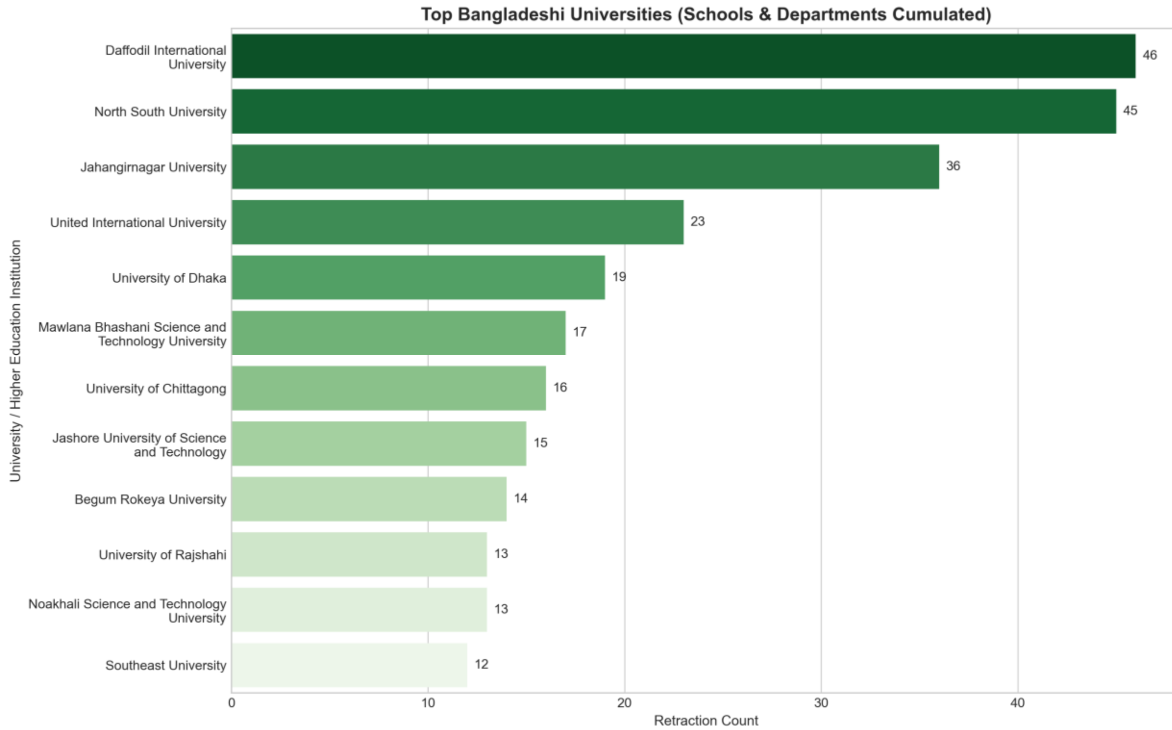

Fig. 13. Top Bangladeshi universities associated with retracted publications (schools and departments normalized).

Description & Findings (Fig. 13): The figure shows that Daffodil International University (46) and North South University (45) have the highest retraction counts among Bangladeshi universities. Jahangirnagar University (36) ranks third, followed by United International University (23) and the University of Dhaka (19). Overall, retractions are concentrated among a small number of institutions, while most universities report substantially lower counts of retractions.

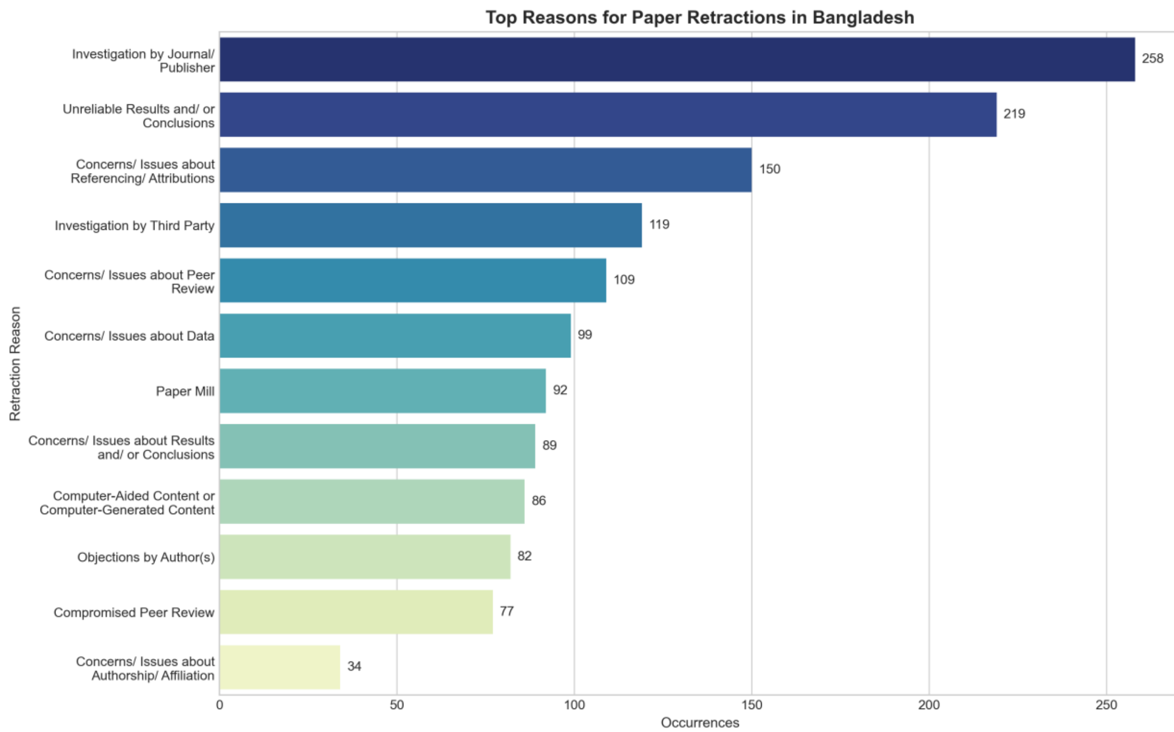

Fig. 14. Top reasons for scholarly paper retractions in Bangladesh.

Description & Findings (Fig. 14): The leading retraction reason in Bangladesh is investigation by the journal/publisher (258 occurrences), followed by unreliable results and/or conclusions (219). Other major factors include concerns about referencing/attribution (150), third-party investigations (119), and peer-review concerns (109). Data-related concerns (99) and paper mill activity (92) also contribute substantially to retractions. Overall, the findings highlight publication integrity, research reliability, and peer-review issues as major drivers of retractions in Bangladesh.

TABLE IX

RETRACTION REASON PREVALENCE: BANGLADESH VS. GLOBAL BASELINE (%)

Reason	Bangladesh (%)	Global (%)	Difference (BD - Global)
Investigation by Journal/Publisher	74.13793103448276	44.11764705882353	30.020283975659236
Unreliable Results and/or Conclusions	62.93103448275862	30.264705882352946	32.66632860040568
Concerns/Issues about Referencing/Attributions	43.103448275862064	21.222689075630253	21.88075920023181
Investigation by Third Party	34.195402298850574	24.665266106442576	9.530136192407998
Concerns/Issues about Peer Review	31.321839080459768	16.264705882352942	15.057133198106824

Reason	Bangladesh (%)	Global (%)	Difference (BD - Global)
Concerns/Issues about Data	28.448275862068968	22.16806722689076	6.280208635178211
Paper Mill	26.436781609195403	16.521008403361346	9.91577320583406
Concerns/Issues about Results and/or Conclusions	25.57471264367816	13.942577030812323	11.632135612865834
Computer-Aided Content or Computer-Generated Content	24.71264367816092	12.84313725490196	11.86950642325896

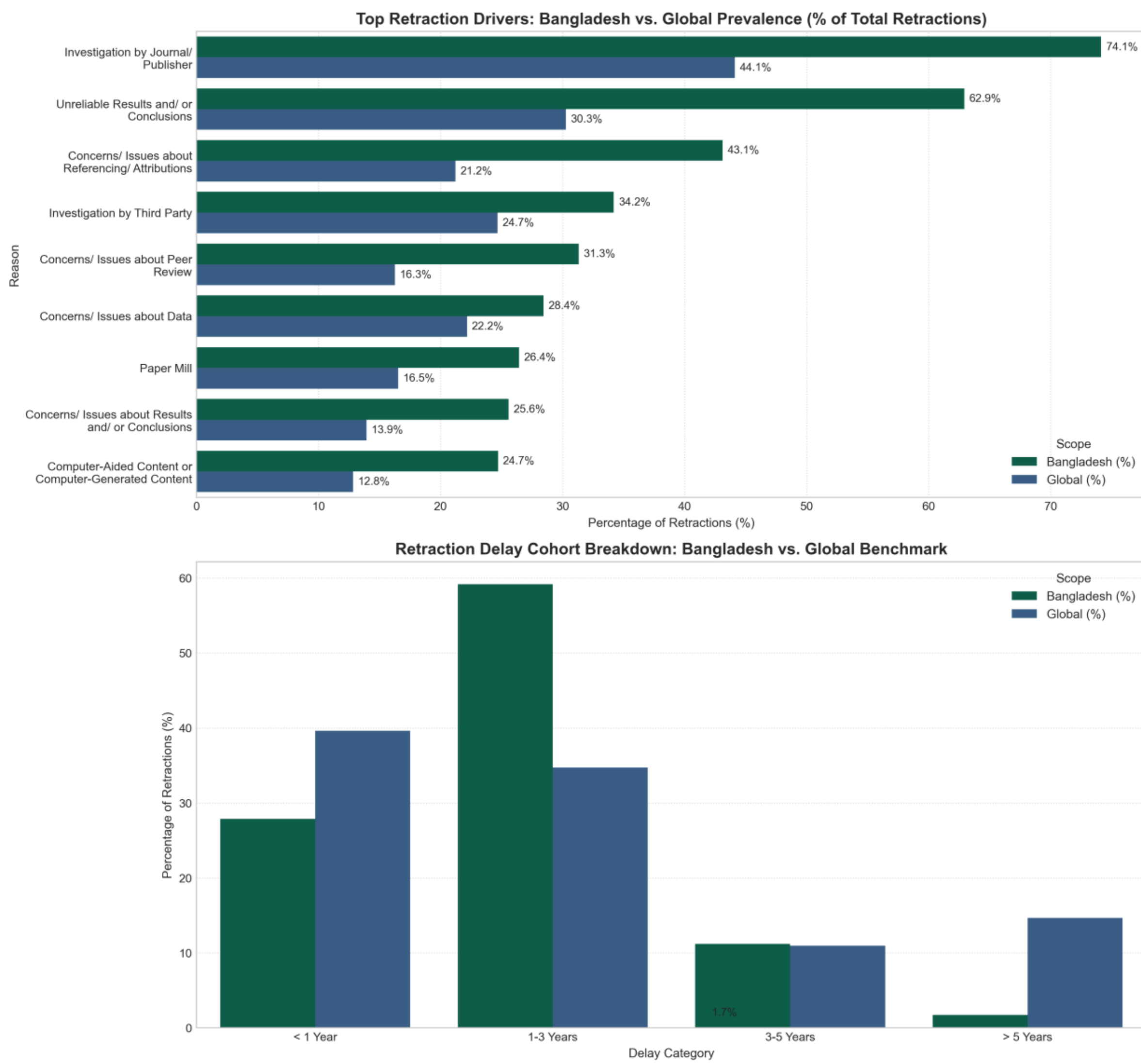

Fig. 15. Direct comparative benchmark: Bangladesh vs. global prevalence (%) and retraction speed cohorts.

Description & Findings (Fig. 15): ***Top Panel:*** Bangladesh exhibits markedly higher prevalence in several key retraction reasons in comparison to global stats, particularly Unreliable Results and/or Conclusions (62.9% vs. 30.3% globally, a +32.7% disparity) and Investigation by Journal/Publisher (74.1% vs. 44.1%, +30.0 points). Concerns/Issues about Referencing/Attributions also show a substantial gap (43.1% vs. 21.2%, +21.9 points), followed by Concerns/Issues about Peer Review (31.3% vs. 16.3%, +15.1 points). ***Bottom Panel:*** The retraction-delay distribution also differs notably: Bangladesh has a much higher proportion of retractions occurring within 1–3 years (59%) compared with the global benchmark (35%), while retractions occurring after 5 years are considerably less common in Bangladesh.

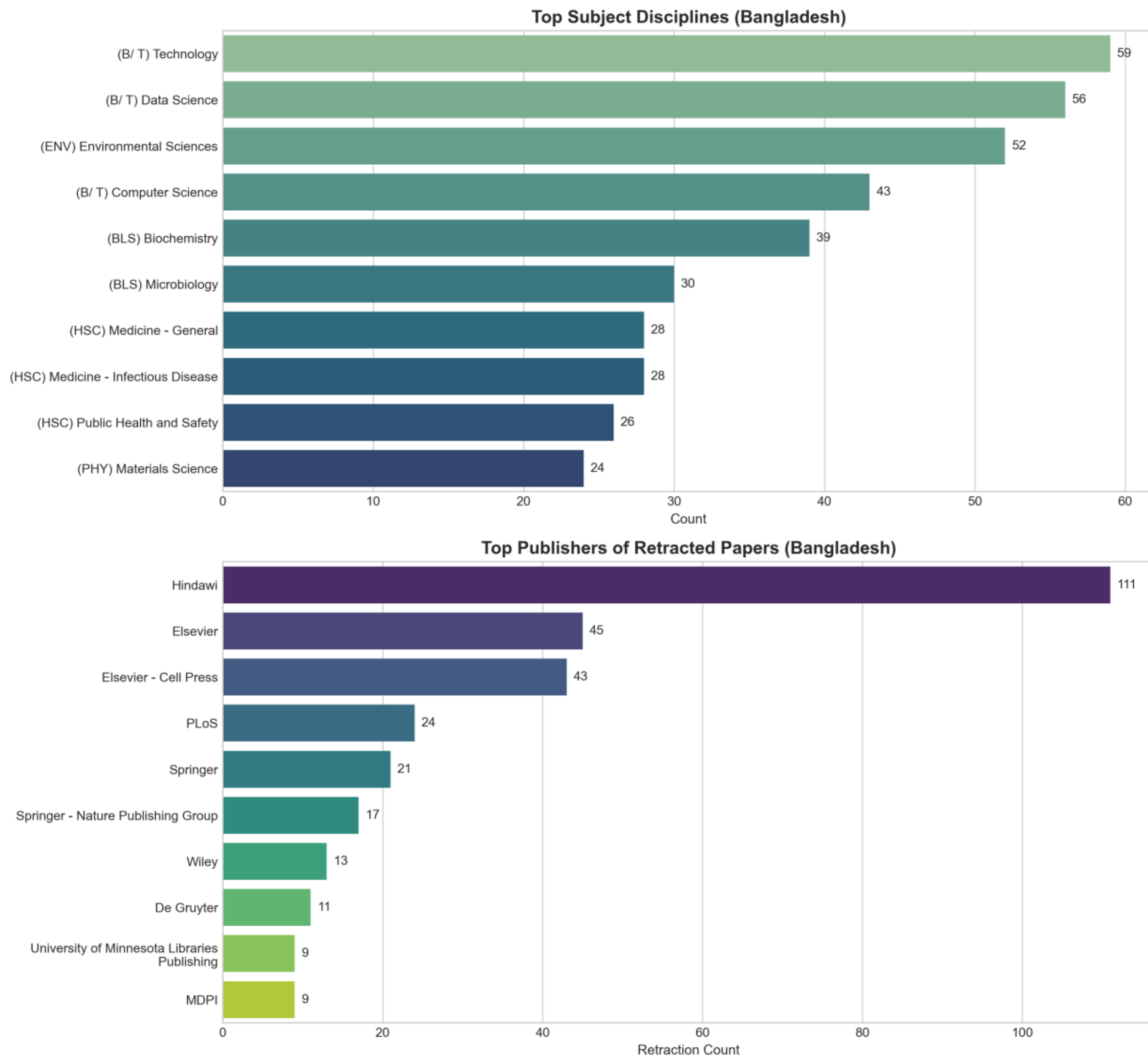

Fig. 16. Top subject disciplines and academic publishers for Bangladeshi retractions.

Description & Findings (Fig. 16): Retractions from Bangladesh are overwhelmingly concentrated in Computer Science, Engineering, Business/Technology, Medicine, and Environmental/Life Sciences, published mainly through IEEE, Hindawi, Springer Nature, and Elsevier.

C. Cross-Border Co-Authorship and International Collaboration Cartels

A critical empirical discovery is that retracted publications involving Bangladesh are not isolated domestic studies. 78.7% of all retracted papers involving Bangladesh represent multi-country cross-border collaborations (only 20.4% are solely domestic) from Fig. 17. Bangladeshi researchers are frequently embedded in international co-authorship networks with researchers in Saudi Arabia (131 co-affiliated retracted papers, 37.6% of total), India (114 papers, 32.8%), Pakistan (53 papers), China (42 papers), Egypt (31 papers), USA (26 papers), and Malaysia (22 papers).

TABLE X

TOP FOREIGN CO-AUTHORING PARTNER NATIONS IN RETRACTED BANGLADESHI PAPERS

Foreign Partner Nation	Co-Affiliated Papers
Saudi Arabia	131
India	114
Pakistan	53
China	42
Egypt	31
United States	26
Malaysia	22
Australia	21
South Korea	20
Brazil	16

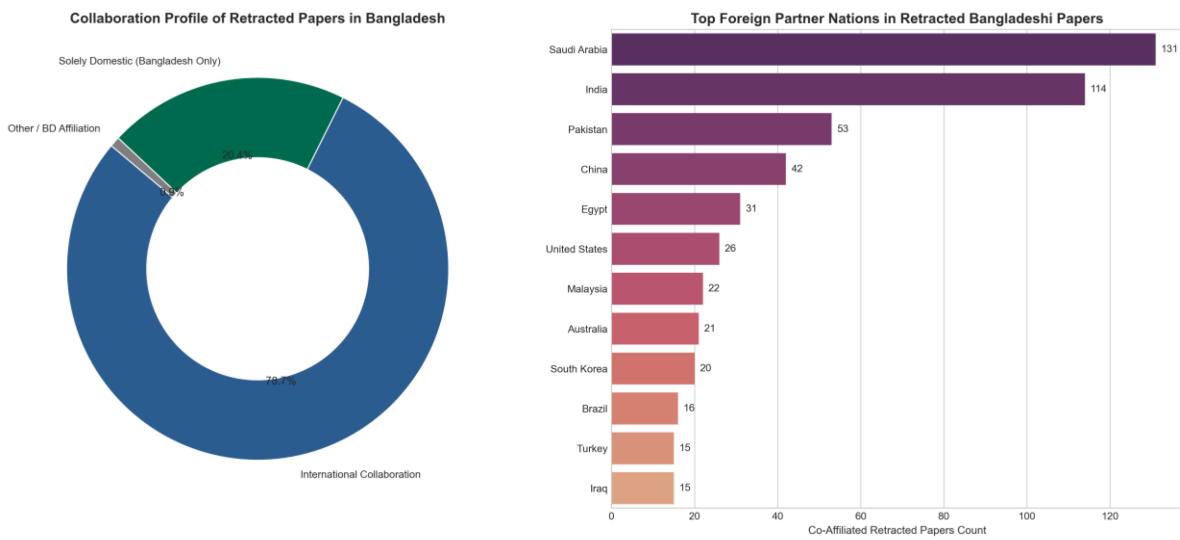

Fig. 17. Collaboration profile (domestic vs. international) and top foreign partner nations in Bangladeshi retractions.

Description & Findings (TABLE X & Fig. 17): Donut chart demonstrates that nearly 4 out of every 5 retracted papers involving Bangladesh (78.7%) are international multi-country co-authorships. Bar chart highlights the heavy concentration of co-affiliations with institutions in Saudi Arabia (131 papers), India (114 papers), Pakistan (53 papers), and China (42 papers), reflecting regional author cartels and shared guest-edited special issue targeting.

D. Hyper-Authorship Inflation and Prolific Repeat-Retracted Authors

Retracted papers involving Bangladesh exhibit severe author team inflation: the average number of authors per retracted paper is 6.72 (median 6.0, maximum 25), compared to the global average of 4.23 authors (median 3.0) from Fig. 18. Furthermore, out of 1,713 distinct authors identified across the dataset, 306 authors (17.9%) have 2 or more retractions, and 35 authors have 5 or more retractions. The top single author alone accounts for 26 retracted papers from TABLE XI.

TABLE XI

TOP PROLIFIC REPEAT-RETRACTED AUTHORS IN BANGLADESH

Author Name	Retractions
Mohammad Monirujjaman Khan	26
Guilherme Malafaia	15
Ahmed Nabih Zaki Rashed	13
Md Habibur Rahman	11

Author Name	Retractions
Md Qamruzzaman	11
Talha Bin Emran	10
A K Mohiuddin	9
Md Mostafizur Rahman	9
Tahia Tazin	9
Sami Bourouis	9

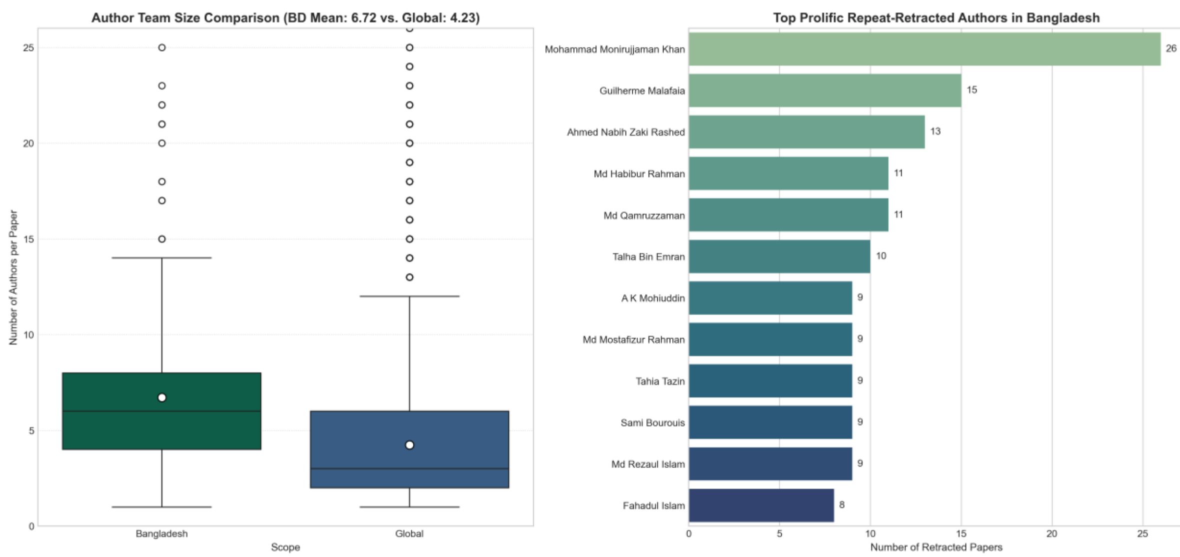

Fig. 18. Author team size distribution comparison (BD vs. Global) and top prolific repeat authors in Bangladesh.

Description & Findings (TABLE XI & Fig. 18): Box plot with mean markers highlights the statistically significant inflation in author team size for Bangladesh (mean 6.72 authors) relative to the global baseline (mean 4.23 authors). Bar chart displays the most prolific repeat-retracted authors, led by Mohammad Monirujjaman Khan (26 retractions), Guilherme Malafaia (15), Ahmed Nabih Zaki Rashed (13), Md Habibur Rahman (11), Md Qamruzzaman (11), and Talha Bin Emran (10).

E. Targeted Scholarly Journals and Institutional Sector Breakdown (Public vs. Private)

Journal-level analysis shows that retractions are heavily concentrated in open-access mega-journals that heavily utilized special issues. Heliyon (Cell Press/Elsevier) accounts for 43 retractions, followed by PLOS ONE (24), BioMed Research International (24), Computational Intelligence and Neuroscience (18), and Science of the Total Environment (16) from TABLE XII. In terms of academic sector, Private universities account for 35.9% of retractions exclusively and participate in 50.6% when joint public-private collaborations (14.7%) are included from TABLE XIII.

TABLE XII

TOP 10 SCHOLARLY JOURNALS RETRACTING RESEARCH INVOLVING BANGLADESH

Journal	Retractions
Heliyon	43
PLoS One	24
BioMed Research International	24
Computational Intelligence and Neuroscience	18
Science of The Total Environment	16
Computational and Mathematical Methods in Medicine	16
Environmental Science and Pollution Research	14
Journal of Healthcare Engineering	12
Journal of Optical Communications	11
INNOVATIONS in pharmacy	9

TABLE XIII

UNIVERSITY SECTOR DISTRIBUTION IN BANGLADESHI RETRACTIONS

Sector	Paper Count
Private Universities	125
Public Universities / Medical	103
Other / Unclassified	69
Joint Public-Private	51

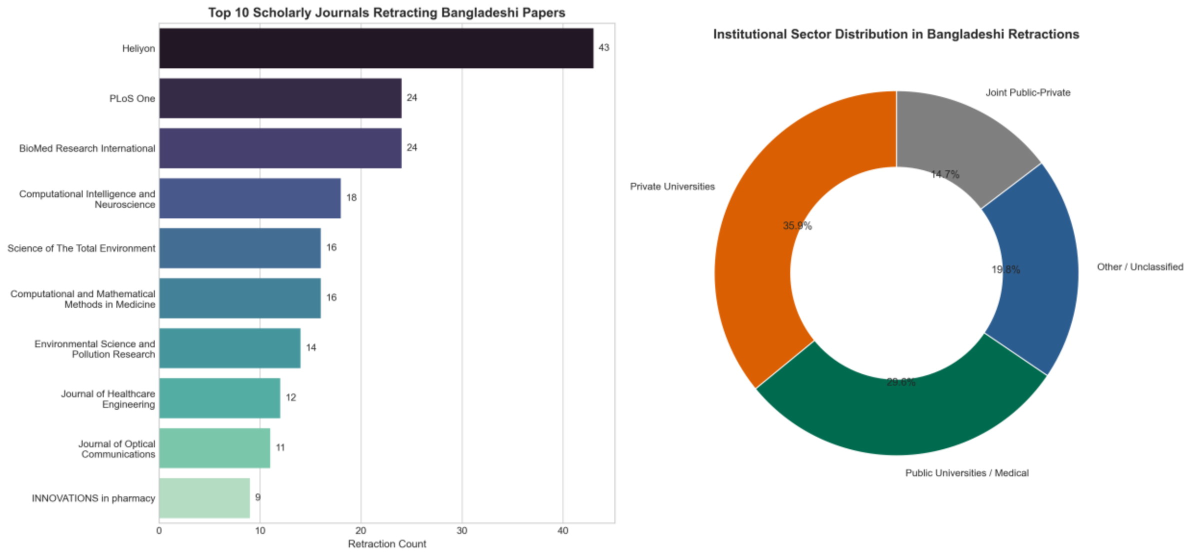

Fig. 19. Top 10 scholarly journals retracting Bangladeshi research and institutional sector breakdown.

Description & Findings (TABLE XII, TABLE XIII & Fig. 19): Bar chart identifies specific journals most impacted by Bangladeshi retractions, led by Heliyon (43 papers), PLOS ONE (24), BioMed Research International (24), and CIN (18). Donut chart details the sector breakdown across Private Universities Only (35.9%), Public / Medical Institutes Only (29.6%), Joint Public-Private Collaborations (14.7%), and others (19.8%).

VIII. CONCLUSIONS AND STRATEGIC RECOMMENDATIONS

Findings from out exploratory research demonstrate retractions in Bangladesh are fueled by structural vulnerabilities in the academic incentive system, international co-authorship cartels, and open-access special issue monetization.

IX. AI USAGE

1. Antigravity was used to convert the original ipynb file to py script.

2. Various AI was used to summarize and rephrase thoughts. Grammar and spelling checks were also done with help of AI in this report.

3. Antigravity was used during the coding phase.

X. REFERENCES

[1] Center for Scientific Integrity, The Retraction Watch Database, New York, NY: The Center for Scientific Integrity, 2024. [Online]. Available: https://retractionwatch.com/

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